

Research Interests

AI for Healthcare, with a focus on **trustworthy multimodal AI** for personalized prediction and clinical decision support. My research integrates wearable and mobile sensing, clinical text, and EHR data to develop uncertainty-aware, interpretable models for personalized healthcare.

Education

Washington University in St. Louis

Ph.D. in Computer Science (AI/ML)

May 2027 (expected)

Advisor: **Prof. Chenyang Lu**, AI for Health Institute

GPA 4.00/4.00

Dissertation: *Trustworthy Multimodal AI for Personalized Healthcare*

B.S. in Computer Science

May 2022

Second Major: Mathematics

GPA 3.99/4.00

Publications

[†] denotes equal contribution or co-first authorship; [‡] denotes co-senior authorship.

First-Author and Equal-Contribution Publications

[P1] **[Neurosurgery 2026]** S. Yakdan[†], **Z. Xu[†]**, J. Zhang, F. Arkam, T. L. Rodebaugh, B. R. Goodin, B. J. Neuman, M. P. Steinmetz, M. P. Kelly, S. Bess, J. Yoon, W. Z. Ray, C. Lu, M. R. Frumkin, J. K. Greenberg. “Real-Time Symptom and Wearable Monitoring May Improve Prediction of Postoperative Recovery in Lumbar Spine Surgery Beyond Standard Measures.” *Neurosurgery*, 2026. doi:10.1227/NEU.0000000000004195

[P2] **[UbiComp 2025]** **Z. Xu**, J. Zhang, S. Haroutounian, H. Liu, Z. Cao, G. R. Messner, H. B. Alaverdyan, S. Ahuja, R. Koshy, J. Hanns, M. Frumkin, T. L. Rodebaugh, C. Lu. “Incorporating Uncertainty in Predictive Models Using Mobile Sensing and Clinical Data: A Case Study on Persistent Post-Surgical Pain.” *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT / UbiComp)*, vol. 9, no. 2, 2025. doi:10.1145/3729488

[P3] **[UbiComp 2024]** **Z. Xu**, J. Zhang, J. K. Greenberg, M. Frumkin, S. Javeed, J. Zhang, B. Benedict, K. Botterbush, T. L. Rodebaugh, W. Z. Ray, C. Lu. “Predicting Multi-Dimensional Surgical Outcomes with Multi-Modal Mobile Sensing.” *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT / UbiComp)*, vol. 8, no. 2, 2024. doi:10.1145/3659628

[P4] **[Neurosurgery 2024]** V. P. Gupta[†], **Z. Xu[†]**, J. K. Greenberg, J. M. Strahle, G. Haller, T. Meehan, A. Roberts, D. D. Limbrick Jr., C. Lu. “Using Artificial Intelligence to Identify Three Presenting Phenotypes of Chiari Type-I Malformation and Syringomyelia.” *Neurosurgery*, 2024. doi:10.1227/neu.0000000000003249

[P5] **[Neurosurgery 2024]** J. K. Greenberg[†], M. R. Frumkin[†], **Z. Xu[†]**, J. Zhang, S. Javeed, J. Zhang, B. Benedict, K. Botterbush, S. Yakdan, C. A. Molina, B. H. Pennicooke, D. Hafez, J. I. Ogunlade, N. Pallotta, M. C. Gupta, J. M. Buchowski, B. Neuman, M. Steinmetz, Z. Ghogawala, M. P. Kelly, B. R. Goodin, J. F. Piccirillo, T. L. Rodebaugh, C. Lu, W. Z. Ray. “Preoperative Mobile Health Data Improve Predictions of Recovery from Lumbar Spine Surgery.” *Neurosurgery*, vol. 5, no. 3, pp. 548–555, 2024. doi:10.1227/neu.0000000000002911

Collaborative Peer-Reviewed Publications

[P6] **[ACL 2026]** S. Wang, **Z. Xu**, C. Najjuuko, C. Alba, C. Lu. “CURA: Clinical Uncertainty Risk Alignment for Language Model-Based Risk Prediction.” *Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026. doi:10.18653/v1/2026.acl-long.1567

[P7] **[ACM TCH 2026]** J. Zhang, R. Wang, **Z. Xu**, H. Liu, J. Rodriguez, H. Cos, R. Srivastava, L. Raper, D. Sanford, C. Hammill, C. Lu. “Addressing Cohort Variability with Adaptive Fusion of Wearable and Clinical Data: A Case Study in Predicting Pancreatic Surgery Outcomes.” *ACM Transactions on Computing for Healthcare*, vol. 7, no. 2, pp. 1–27, 2026. doi:10.1145/3788681

[P8] **[BMJ Open 2026]** K. J. Holzer, H. Alaverdyan, **Z. Xu**, M. R. Frumkin, K. A. Frey, S. H. Gregory, T. L. Rodebaugh, C. Lu, C. R. King, D. Head. “Protocol for Personalised Prediction of Persistent Postsurgical Pain.” *BMJ Open*, vol. 16, no. 2, e107055, 2026. doi:10.1136/bmjopen-2025-107055

[P9] **[Br. J. Anaesth.]** M. R. Frumkin, G. R. Messner, K. J. Holzer, **Z. Xu**, T. L. Rodebaugh, H. Bernstein, K. Frey, C. Lu, S. Haroutounian. “Remote Perioperative Symptom Monitoring via Smartphone is Feasible: Evidence from the Personalized Prediction of Persistent Postsurgical Pain (P5) Study.” *British Journal of Anaesthesia*, accepted.

[P10] **[ACL Findings 2025]** B. C. Warner, **Z. Xu**, S. Haroutounian, T. Kannampallil, C. Lu. “Utilizing Semantic Textual Similarity for Clinical Survey Data Feature Selection.” *Findings of the Association for Computational Linguistics*

(ACL), pp. 502–520, 2025. doi:10.18653/v1/2025.findings-acl.27

[P11] [Nat. Commun. 2025] C. Najjuuko, Z. Xu, S. Kizito, C. Lu, F. M. Ssewamala. “Patterns of Maternal and Child Health Services Utilization and Associated Socioeconomic Disparities in Sub-Saharan Africa.” *Nature Communications*, vol. 16, 7840, 2025. doi:10.1038/s41467-025-61350-8

[P12] [AIDS 2025] C. Najjuuko, R. Brathwaite, Z. Xu, S. Kizito, C. Lu, F. M. Ssewamala. “Using Machine Learning to Predict Poor Adherence to Antiretroviral Therapy Among Adolescents Living with HIV in Low-Resource Settings.” *AIDS*, vol. 39, no. 9, pp. 1204–1213, 2025. doi:10.1097/QAD.0000000000004163

[P13] [J. Pain 2025] M. R. Frumkin, J. Zhang, Z. Xu, S. Yakdan, B. Benedict, S. Javeed, J. Zhang, K. Botterbush, B. R. Goodin, C. Lu, W. Z. Ray, J. K. Greenberg. “Rethinking Measurement of Movement-Evoked Pain with Digital Technology.” *The Journal of Pain*, 105643, 2025. doi:10.1016/j.jpain.2025.105643

[P14] [Spine J. 2025] S. Yakdan, J. Zhang, B. Benedict, Z. Xu, S. Javeed, J. K. Zhang, B. A. Steel, V. P. Gupta, K. Botterbush, J. F. Piccirillo, T. L. Rodebaugh, B. R. Goodin, J. M. Buchowski, B. Neuman, D. Hafez, M. P. Kelly, W. Z. Ray, C. Lu, M. R. Frumkin, J. K. Greenberg. “Multi-Domain Postoperative Recovery Trajectories After Lumbar and Thoracolumbar Spine Surgery.” *The Spine Journal*, 2025. doi:10.1016/j.spinee.2025.05.017

[P15] [Spine Deformity 2025] R. Bhan, S. Yakdan, C. Diaz, J. Zhang, Z. Xu, K. Joseph, B. Plog, A. T. Yahanda, J. K. Zhang, S. Javeed, W. Z. Ray, M. Gupta, C. Lu, M. P. Kelly, J. K. Greenberg, B. Neuman. “Association Between Preoperative Activity Levels and Postoperative Complications in Adult Spinal Deformity Surgery.” *Spine Deformity*, 2025. doi:10.1007/s43390-025-01158-0

[P16] [Clin. Spine Surg. 2025] S. Yakdan, J. Zhang, K. Botterbush, M. I. Kaleem, B. Benedict, Z. Xu, M. R. Frumkin, S. Javeed, J. K. Zhang, J. Ogunlade, C. Lu, B. Neuman, M. P. Steinmetz, W. Z. Ray, M. Gupta, J. K. Greenberg. “Sleep Physiology in Spine Disease: A Comparative Study of Cervical and Lumbar Patients.” *Clinical Spine Surgery*, 2025. doi:10.1097/BSD.0000000000001926

[P17] [KDD 2023] B. Xue, A. Said, Z. Xu, H. Liu, N. Shah, H. Yang, P. Payne, C. Lu. “Assisting Clinical Decisions for Scarcely Available Treatment via Disentangled Latent Representation.” *ACM SIGKDD Conference on Knowledge Discovery & Data Mining (KDD)*, 2023. doi:10.1145/3580305.3599774

[P18] [Artif. Organs 2023] N. Shah, B. Xue, Z. Xu, H. Yang, E. Marwali, H. Dalton, P. P. R. Payne, C. Lu, A. S. Said, ISARIC Clinical Characterisation Group. “Validation of Extracorporeal Membrane Oxygenation Mortality Prediction & Severity of Illness Scores in an International COVID-19 Cohort.” *Artificial Organs*, vol. 47, pp. 1490–1502, 2023. doi:10.1111/aor.14542

[P19] [JNS Spine 2023] J. K. Greenberg, S. Javeed, J. K. Zhang, B. Benedict, M. R. Frumkin, Z. Xu, J. Zhang, T. L. Rodebaugh, C. Lu, M. Steinmetz, Z. Ghogawala, M. Bydon, W. Z. Ray. “Current and Future Applications of Mobile Health Technology for Evaluating Spine Surgery Patients: A Review.” *Journal of Neurosurgery: Spine*, 2023. doi:10.3171/2022.12.SPINE221302

Manuscripts Under Review

[X1] [Neurosurgery] Z. Xu, W. Mualem, S. Venigalla, J. K. Greenberg, T. Meehan, J. M. Strahle, J. L. Roland, G. Haller, D. D. Limbrick Jr., C. Lu. “Expert Opinion or Data-Driven Features? AI-Based Prediction of Headache Outcomes from Surgery for Chiari 1 Malformation with Syringomyelia.” *Neurosurgery*, under review, 2026. Corresponding author.

[X2] [J. Alzheimer’s Dis.] K. Patel[†], Z. Xu[†], M. Dolatshahi, P. Gao, L. Lloyd, P. K. Commean, C. Lu, C. A. Raji. “Machine Learning Prediction of Amyloid Accumulation Linked to Adiposity for Alzheimer’s Disease Assessment.” *Journal of Alzheimer’s Disease*, under major revision, 2026.

Conference Abstracts & Review Articles

[A1] [Neurosurgery 2026] S. Yakdan[†], Z. Xu[†], J. Zhang, J. F. Piccirillo, T. L. Rodebaugh, G. Burel, B. J. Neuman, M. P. Steinmetz, W. Z. Ray, C. Lu. “Leveraging Preoperative Mobile Health Data to Predict 1-Year Recovery Outcomes After Lumbar Spine Surgery.” *Neurosurgery*, vol. 72, suppl. 1, p. 43, 2026. doi:10.1227/neu.0000000000003964_202

[A2] [Neurosurgery 2025] J. Zhang, S. Yakdan, B. C. Benedict, J. Zhang, M. I. Kaleem, S. Javeed, K. S. Botterbush, M. R. Frumkin, Z. Xu, C. Lu. “Minimum Clinically Important Differences in Quantitative Objective Markers of Physical Function in Patients Undergoing Lumbar Spine Surgery.” *Neurosurgery*, vol. 71, suppl. 1, pp. 99–100, 2025. doi:10.1227/neu.0000000000003360_414

[A3] [Semin. Spine Surg. 2023] S. Yakdan, K. Botterbush, Z. Xu, C. Lu, W. Z. Ray, J. K. Greenberg. “Machine Learning and Lumbar Spondylolisthesis.” *Seminars in Spine Surgery*, vol. 35, no. 3, 101048, 2023. doi:10.1016/j.semss.2023.101048

Experience

Research Assistant, Washington University in St. Louis

2022 – Present

Advisor: *Prof. Chenyang Lu*, AI for Health Institute

- **Trustworthy multimodal AI for personalized healthcare.** Led research on uncertainty-aware and multitask learning across wearable sensing, ecological momentary assessment (EMA), and clinical data for personalized prediction of postsurgical pain and recovery. Published in [P1], [P2], and [P3].
- **Clinical AI and language models.** Investigated trustworthy language-model-based clinical prediction using semantic representation, parameter-efficient fine-tuning, uncertainty quantification, and confidence-aware human review. Published in [P6] and [P10].
- **Interpretable clinical modeling.** Developed clinician-informed and data-driven phenotyping and outcome-prediction methods for neurosurgical populations, connecting machine-learning methodology with clinically meaningful patient stratification. Published in [P4]; related work under review in [X1].
- **Translational AI for health.** Collaborated with Anesthesiology, Neurosurgery, Pain Medicine, and Public Health researchers to translate clinical questions into machine-learning studies with clinically meaningful evaluation and interpretation.

Applied AI Agents Intern, Apple

May 2026 – Aug 2026

- Developed and evaluated AI-agent systems for scientific and engineering workflows using multimodal retrieval, tool use, layered memory, simulation control, and uncertainty-aware evaluation; achieved 96% retrieval success across a 130-question evaluation set.

Grant and Sponsored Research Experience

Funded Research Award

Leveraging Machine Learning and Wearable Device Activity Patterns to Improve Prehabilitation and Surgical Outcomes, BJC HealthCare Innovation Lab & WashU

2025–2026

I2DB Big Ideas Competition; \$50,000; Team Leads: D. Sanford and C. Lu

Team member. Developed wearable-activity prediction models and contributed the corresponding methods and results to the funded proposal.

Contributions to Funded Grant Proposals

Contributed predictive modeling, preliminary analyses, and technical writing to the following funded proposals; not named personnel on these awards.

IMPETUS: Integrated Mechanisms, Phenotypes, and Translational Underpinnings of Chronic Pain after Surgery, NIH NINDS RM1NS135283; MPI: Haroutounian,

2024

Creed, Lu, Rodebaugh, Shepherd, Sinha

Developed multimodal machine-learning models for chronic postsurgical pain phenotyping and contributed the associated methods and preliminary results to the proposal.

Redefining Chiari Type I Malformation and Its Impact on Brain Development, NIH

2023

NINDS P01NS131131; MPI: D. D. Limbrick Jr. and C. Lu

Developed phenotyping and outcome-prediction models and contributed the corresponding methods and preliminary results to the proposal.

Invited Talks and Selected Presentations

Invited Talks

“Beyond Prediction: Trustworthy Multimodal AI for Pain.”

2026

#Pain2026 AI and Pain Workshop, Gulf Coast Consortia

“Using AI to Predict Post-Surgical Headache Improvement for Chiari I and Syringomyelia.”

2026

Park-Reeves Chiari & Syringomyelia Research Center Annual Update

“Expert Opinion or Data-Driven Features? AI-Based Prediction of Headache Outcomes from Surgery for Chiari I Malformation with Syringomyelia.”

2026

Bobby Jones Chiari & Syringomyelia Foundation Think Tank

Conference Presentations

- “Incorporating Uncertainty in Predictive Models Using Mobile Sensing and Clinical Data: A Case Study on Persistent Post-Surgical Pain.” 2025
ACM UbiComp/IMWUT, Helsinki, Finland. [P2]
“Predicting Multi-Dimensional Surgical Outcomes with Multi-Modal Mobile Sensing.” 2024
ACM UbiComp/IMWUT, Melbourne, Australia. [P3]

Teaching Experience

- Guest Lecturer**, CSE 419A: *Introduction to AI for Health* Fall 2025, Fall 2026
Guest Lecturer, PHS 5100: *Development, Validation and Application of Risk Prediction* Spring 2026
Assistant Instructor, CSE 419A: *Introduction to AI for Health* Fall 2024
Assistant Instructor, CSE 531A: *AI for Health* (graduate) Spring 2024
Teaching Assistant, CSE 417: *Introduction to Machine Learning* Spring 2023
Teaching Assistant, CSE 240: *Logic and Discrete Mathematics* Spring 2020

Student Research Mentoring

Mentored eight students (one Ph.D. student, one Ph.D. rotation student, one M.S. student, four undergraduate students, and one medical student) in multimodal sensing, uncertainty quantification, and clinical machine learning, in collaboration with Prof. Chenyang Lu. **Five mentees co-authored publications, including three as first or co-first authors**; two received institutional research or academic awards.

Ph.D. and Graduate Students

- Claire Najjuuko**, Ph.D. student; co-advised by Prof. Chenyang Lu and Prof. F. M. Ssewamala 2023–Present
Mentored machine-learning components of dissertation research on maternal–child health and antiretroviral therapy adherence in Sub-Saharan Africa. **First author** on [P11] (*Nature Communications*) and [P12] (*AIDS*); recipient of the 2025 Provost’s Research Excellence Award.
Sizhe Wang, Ph.D. rotation student, Computer Science 2025
Mentored research on clinical uncertainty–risk alignment for language-model-based risk prediction. **First author** on [P6] (*ACL* 2026). Now a Ph.D. student at Washington University in St. Louis.
Yuyang Wu, M.S. Computer Science 2025
Mentored research on uncertainty quantification for language-model-based clinical prediction. Master’s thesis: *Quantifying Uncertainty in Clinical Language Models: When and How It Is Useful for Clinical Prediction*. Now Machine Learning Engineer at TikTok.

Undergraduate and Medical Students

- Lucien Liu**, B.S. Computer Science 2026
Mentored rhythm-based feature extraction from ecological momentary assessment (EMA) time series. Presented at the WashU Undergraduate Research Symposium; manuscript in preparation. Now M.S. student at Massachusetts Institute of Technology.
Kavya Patel, B.S. Computer Science; Minor in Chemistry 2025
Mentored machine-learning modeling of amyloid accumulation and its association with adiposity in Alzheimer’s disease. **Co-first author** on [X2], under major revision at *Journal of Alzheimer’s Disease*. Now Associate Consultant at Bain & Company.
Siddharth Venigalla, M.D. candidate 2025
Mentored data processing, experimental design, and interpretation for prediction of headache outcomes after surgery for Chiari I malformation with syringomyelia. Co-author on [X1], under review at *Neurosurgery*.
Tongyang (Jason) Wang, B.S. Computer Science 2025–Present
Mentoring wearable-sensing research for prediction of adverse postsurgical outcomes in the SPAR cohort; manuscript in preparation. Recipient, Outstanding Junior Award for Academic Excellence, WashU Department of Computer Science & Engineering, 2026.
Zihan Cao, B.S. Computer Science 2023
Mentored uncertainty-aware multimodal modeling for persistent postsurgical pain prediction. Co-author on [P2] (*ACM IMWUT/UbiComp* 2025). Now M.S. student at Duke University.

Honors and Awards

WashU Big Ideas Competition – \$50K Research Award (team member)	2025
Doctoral Honors – Top 15–20% in WashU’s Periodic Review of Doctoral Students	2024–2026
Outstanding Reviewer – <i>Journal of Digital Health</i>	2025
Dean’s Select Ph.D. Fellowship – Washington University in St. Louis	2022–2027
Summa Cum Laude – Washington University in St. Louis	2022

Professional Service

Program Committees

- Program Committee, ACM CIKM Workshop on Large-scale AI for Sensing Systems (LASS), 2026.
- Program Committee, International Multi-Conference on Artificial Intelligence Technology, 2026.

Peer Review

- **Conference reviewing:** ACL Rolling Review (2026).
- **Journal reviewing:** *Nature Communications* (early-career co-reviewer); *ACM Transactions on Computing for Healthcare* (2026); *Scientific Reports* (2024–2026); *BMC Medical Informatics and Decision Making* (2025–2026); *BMC Anesthesiology* (2026); *Journal of Digital Health* (2025).

Departmental and Community Service

- **Graduate Student Advisory Board Member**, WashU Center for Career Engagement, 2026–2027.
- **Volunteer Organizer**, WashU AI for Health Symposium (2024–2025) and AI for Health Venture Day (2026).
- **Test Writer & Tournament Volunteer**, WashU Science Olympiad, 2026–2027. Develop and proctor STEM competition events supporting the WashU Invitational (1,800+ middle- and high-school students) and Missouri Regional Tournament.